PASSES-HGV: Expanding the Continuous Spectral Framework to 6-DOF Hypersonic Glide and Fractional Orbital Trajectories

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Abstract

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1 Introduction

2 The Expanded Global State Vector

To handle skipping trajectories, waverider aerodynamics, and orbital mechanics, the vehicle can no longer be treated as a purely axisymmetric point-mass with 1D bending. The global state vector $\mathbf{X}_{global}$ must be expanded to full 6-Degree-of-Freedom (6-DOF) rigid body kinematics.

- **Attitude Representation:** We will need to introduce a quaternion vector $\mathbf{q} = [q_0, q_1, q_2, q_3]^T$ into the state to avoid gimbal lock during extreme atmospheric maneuvering and partial-orbit insertion.
- **The New Vector:** The state expands from $[\mathbf{r}_E, \mathbf{v}_E, m_{bulk}, \mathbf{w}, \dot{\mathbf{w}}, \mathbf{H}]^T$ to include the quaternion $\mathbf{q}$ and the body-frame angular velocity $\boldsymbol{\omega}_B$.

3 Unclassified Hypersonic Aerothermodynamics

Since we do not have access to classified CFD databases for specific waverider geometries, we can mathematically close the aerodynamic and thermodynamic loops using well-established, unclassified hypersonic theories.

- **Modified Newtonian Aerodynamics:** For extreme Mach numbers, the localized pressure coefficient on any structural node can be rigorously approximated using Newtonian flow theory:
$$C_p = C_{p,max} \sin^2 \delta_c$$
where δ_c is the local impact angle of the velocity vector relative to the spectral surface normal. This allows you to generate realistic, continuous spatial pressure distributions $\mathbf{Q}_{aero}$ across the vehicle body without needing look-up tables.
- **Sutton-Graves Convective Heating:** For the continuous thermal ablation boundary condition, we can utilize the classic empirical Sutton-Graves equation for stagnation point heat flux:
$$\dot{q}_{conv} = k\sqrt{\rho}V_r^3$$

This provides a violently non-linear, highly realistic heat flux forcing function to feed into your smoothed Enthalpy Method, perfectly stressing the RKF45 solver during a simulated 20-minute glide phase.

4 Structural Generality: Beyond Euler-Bernoulli

HGVs are lifting bodies. They do not just bend along a 1D longitudinal axis; they experience severe torsional (twisting) forces and 3D aeroelastic flutter due to continuous bank-angle modulation and asymmetric aerodynamic loading.

- **Spectral Plate Expansion:** We can elevate the structural RHS matrix from the 1D Euler-Bernoulli beam theory to a 2D Kirchhoff-Love or Mindlin plate theory, still utilizing the Chebyshev Collocation philosophy. By mapping the continuous structural domain to a bivariate spectral grid $\xi, \eta \in [-1, 1]$, you preserve the exact C^∞ geometric transcription you already established, while unlocking multi-dimensional aeroelastic modes.

5 GNC: Energy Management vs. Terminal Intercept

AC-APN is useless for a 20-minute boost-glide trajectory. The HGV is effectively a glider trading kinetic energy for range, bounded by strict thermal and structural limits.

- **Predictor-Corrector Guidance:** The GNC section will shift from APN to a numerical Predictor-Corrector algorithm. Since your MOL physics engine is completely vectorized and GPU-accelerated, the flight computer can literally call a low-resolution "shadow" version of the PASSES physics engine in real-time. It can rapidly project the current state forward to predict the footprint, and iteratively adjust the bank-angle command to bleed off energy while staying within the structural heat-flux boundary.

6 Orbital Mechanics and FOBS

To capture the FOBS (Fractional Orbital Bombardment System) requirement, we implement a long-duration exo-atmospheric coast phase.

- **The Physics Idle:** When the dynamic pressure $q \to 0$, the aerodynamic forcing and convective heating naturally zero out in your continuous equations. The structural and thermal solvers effectively "idle" computationally, while the 6-DOF kinematics switch to Keplerian orbital propagation governed strictly by the $J_2 - J_4$ gravitational gradients until the de-orbit burn is commanded.

7 Conclusion